Flashcards — Biology Chapter 7: Metabolism

How to use these cards: Cover the right column with your hand or a piece of paper. Read the question, say your answer out loud, then slide down to check. If you get it wrong, put a dot next to it — come back and re-test the dots at the end.

Daily target: 15 minutes of flashcards each day. Mix the order so your brain has to work harder.

Cards are organized by **topic** and **difficulty:**
- **Easy** — facts and definitions (you should get 100%)
- **Medium** — apply a rule or explain a concept
- **Hard** — multi-step thinking, comparisons, or tricky distinctions

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Topic 1 — Metabolism and Its Types

Easy

Q	A
What does the word "metabolism" mean in Greek?	"Change"
Define metabolism.	The sum of all chemical reactions occurring within each cell of a living organism for the maintenance and sustainability of life
What are the two types of metabolism?	Catabolism and anabolism
Define catabolism.	Biochemical reactions that break down complex molecules into simpler ones, releasing energy
Define anabolism.	Biochemical reactions that build complex molecules from simpler ones, consuming energy
Is digestion an example of catabolism or anabolism?	Catabolism
Is photosynthesis an example of catabolism or anabolism?	Anabolism

Medium

Q	A
Give one example of a catabolic reaction.	Cellular respiration (glucose → CO_2 + H_2O) OR digestion of proteins → amino acids
Give one example of an anabolic reaction.	Photosynthesis (CO_2 + H_2O → glucose) OR glycogenesis (glucose → glycogen)
In which type of metabolic reaction is energy released?	Catabolism
In which type of metabolic reaction is energy utilized?	Anabolism
What is the relationship between catabolism and anabolism in the body?	Energy released in catabolism is utilized in anabolism; they work together

Hard

Q	A
Cellular respiration breaks glucose into CO_2 and H_2O. Why is this catabolism and not just "breathing"?	Breathing is the physical act; respiration is the biochemical process inside cells where complex glucose is broken into simpler products, releasing energy — the definition of catabolism
Why must catabolism and anabolism both	Catabolism provides energy; anabolism uses that energy to build and maintain

occur simultaneously in a living organism?	cell structures. Stopping either would cause the organism to die — from energy starvation or structural collapse

Topic 2 — Enzymes and Their Characteristics

Easy

Q	A
What are enzymes?	Biologically active globular proteins made by living cells that tremendously speed up biochemical reactions
Another name for enzymes?	Biological catalysts
What is the study of enzymes called?	Enzymology
What are substrates?	The molecules that enzymes act on
What are products (in enzyme context)?	The substrates modified by enzymes
Name the two types of enzymes based on location.	Intracellular enzymes and extracellular enzymes
Give an example of an intracellular enzyme.	Enzymes of glycolysis (in cytoplasm)
Give an example of an extracellular enzyme.	Pepsin (works in stomach cavity)
Are all enzymes proteins?	Most are; a few are RNA molecules called ribozymes
What is a cofactor?	A non-protein component required for proper enzyme functioning

Medium

Q	A
Name the three types of cofactors.	Activators, prosthetic groups, and coenzymes
What is an activator (cofactor)?	Detachable inorganic ions that help enzyme activity, e.g., Zn^{2+}, Fe^{2+}, Cl^-
What is a prosthetic group?	Organic molecule tightly and permanently attached to an enzyme, e.g., FAD, Haem
What is a coenzyme?	Detachable organic molecule that assists enzymes, e.g., NAD, Coenzyme A, Vitamins A and C
Why can a small quantity of enzyme catalyze a large amount of substrate?	Enzymes regain their original shape after each reaction and are reused repeatedly
What does it mean that enzymes are "specific"?	Each enzyme catalyzes only one type of reaction and will not act on different substrates
Give an example of enzyme specificity.	Amylase acts only on starch, not on proteins or fats
What is the difference between an enzyme and an activator?	Enzyme is the protein catalyst; activator is a cofactor (inorganic ion) that helps the enzyme function

Hard

Q	A
Why would a vitamin deficiency impair enzyme activity?	Many vitamins function as coenzymes; without them, enzymes lack their cofactor and cannot function properly
An enzyme lowers activation energy but cannot start an impossible reaction. Explain this distinction.	Activation energy is the barrier to a possible reaction; enzymes lower this barrier. But if the products are thermodynamically less stable than reactants, no amount of enzyme can drive the reaction — it's impossible regardless

Why are ribozymes significant?	They show that proteins are not the only molecules that can catalyze biochemical reactions; RNA can also act as a catalyst, important for theories of early life

Topic 3 — Mechanism of Enzyme Action

Easy

Q	A
What is the active site of an enzyme?	A small, charge-bearing cavity on the enzyme surface where the substrate binds and the reaction takes place
What is the ES complex?	Enzyme-Substrate Complex — formed when substrate binds to the enzyme's active site
What is the EP complex?	Enzyme-Product Complex — the ES complex after the reaction has occurred
What happens to the enzyme at the end of the reaction?	It is released unchanged and available for the next reaction
What is the lock-and-key model of enzyme action?	The enzyme (lock) has an active site that is complementary in shape to the substrate (key); only the right substrate fits
What is activation energy?	The minimum energy required to start a chemical reaction
How do enzymes affect activation energy?	They lower it, allowing reactions to occur at body temperature

Medium

Q	A
Write the sequence of enzyme action steps.	Enzyme + Substrate → ES Complex → EP Complex → Products + Enzyme
What is the induced fit model?	The enzyme's active site is flexible and adjusts its shape when the substrate binds, gripping it more tightly
How is the induced fit model different from the lock-and-key model?	Lock-and-key assumes a rigid, fixed active site; induced fit recognizes that the enzyme slightly changes shape to accommodate the substrate
Why does denaturation destroy enzyme function?	It destroys the 3D shape of the active site, so the substrate can no longer bind
By changing the shape and charge of the substrate, enzymes lower activation energy. Explain.	By binding the substrate and distorting it slightly, the enzyme makes the substrate more reactive — easier to convert — reducing the energy barrier

Hard

Q	A
The active site is a very small portion of the enzyme. Why is most of the enzyme not involved in catalysis?	The rest of the enzyme provides the structural scaffold that gives the active site its correct 3D shape, charge, and stability. Without the rest of the protein, the active site couldn't hold its functional shape
An enzyme shows two substrates binding at the active site (synthesis reaction). How is this possible if active sites are specific?	The active site can be shaped to accommodate two specific molecules simultaneously — and bring them together to form one product. Specificity means only the right pair can bind, not that only one molecule can bind

Topic 4 — Factors Affecting Enzyme Activity

Easy

Q	A
Name the three main factors that affect enzyme activity.	Temperature, pH, and substrate concentration
What is optimum temperature?	The temperature at which an enzyme works at its maximum rate
What is the optimum temperature for human enzymes?	36°C–38°C (approximately 37°C)
What happens to an enzyme at very high temperatures?	It is denatured — the active site is destroyed and the enzyme can no longer function
What is denaturation?	Irreversible destruction of an enzyme's 3D structure (active site) due to extreme heat or pH
What is optimum pH?	The pH at which an enzyme shows maximum activity
What is the optimum pH for pepsin?	pH 2.0 (acidic)
What is the optimum pH for trypsin?	pH 8.0 (alkaline)
What happens to enzyme activity at very low temperatures?	It becomes inactive due to insufficient activation energy; activity returns when temperature is restored

Medium

Q	A
What is saturation of the active site?	When all enzyme active sites are occupied by substrate, so further increases in substrate concentration have no effect on reaction rate
What happens to reaction rate as substrate concentration increases, all else equal?	Rate increases proportionally until all active sites are occupied (saturation), then plateaus
Pepsin works in the stomach but not the intestine. Explain in terms of pH.	Pepsin has an optimum pH of 2 (acidic); the intestinal environment is alkaline (~pH 8), so pepsin is inactive or denatured there
If you cool an enzyme to 0°C, then warm it back to 37°C, will it work?	Yes — cold inactivates but does not denature enzymes; activity returns when temperature is restored
What are inhibitors and activators?	Inhibitors are molecules that stop/slow enzymatic reactions; activators are molecules that increase enzymatic reaction rate

Hard

Q	A
Why does the optimum temperature of human enzymes (37°C) match normal body temperature?	Evolution has tuned enzymes over millions of years to work best at the organism's normal body temperature — it would be inefficient otherwise
A substrate concentration graph plateaus. What structural explanation accounts for the plateau?	All enzyme active sites are occupied (saturated). The reaction rate is now limited by the number of enzyme molecules, not substrate. Adding more substrate simply queues up, waiting for an active site to become free
Why does changing pH alter enzyme activity even without high temperatures?	pH changes alter the ionization state of amino acids in the active site, changing its charge distribution and shape — the substrate can no longer bind properly even though the protein hasn't been physically destroyed by heat

Topic 5 — Enzyme Inhibition

Easy

Q	A
Define enzyme inhibition.	The phenomenon where enzymes cannot perform their role due to interference by

	chemicals at the reaction site
What is an inhibitor?	A chemical that reacts with the enzyme in place of the substrate but does not convert to products, blocking enzyme action
Name the two types of enzyme inhibition.	Competitive inhibition and non-competitive inhibition
Where does a competitive inhibitor bind?	At the active site
Where does a non-competitive inhibitor bind?	At the allosteric site (different from the active site)
What is the allosteric site?	A site on the enzyme surface, other than the active site, where non-competitive inhibitors bind

Medium

Q	A
How does competitive inhibition work?	The inhibitor is structurally similar to the substrate and occupies the active site, preventing the real substrate from binding
How does non-competitive inhibition work?	The inhibitor binds to the allosteric site, changing the enzyme's shape and damaging the active site, so the substrate cannot bind properly
Is competitive inhibition reversible? How?	Yes — by increasing the concentration of the real substrate, which out-competes the inhibitor for the active site
Why is non-competitive inhibition generally irreversible?	The inhibitor permanently distorts the enzyme's 3D shape; the active site is damaged even if the inhibitor is later removed
Give an example of a real-world competitive inhibitor.	Aspirin inhibits COX enzyme competitively; some drugs mimic substrates to block enzymes

Hard

Q	A
How does cyanide kill?	Cyanide is a non-competitive inhibitor of cytochrome c oxidase (final enzyme in the ETC). This stops ATP production in all cells simultaneously, causing rapid death
How does penicillin kill bacteria but not human cells?	Penicillin inhibits the enzyme bacteria use to build cell walls. Human cells don't have this type of cell wall or enzyme, so penicillin specifically targets bacterial metabolism
If a cell produces too much of a metabolic product, it might inhibit the first enzyme in the pathway. What type of inhibition is this, and why is it useful?	This is feedback inhibition (usually non-competitive). It is useful because it automatically regulates production — when enough product exists, it shuts off the pathway. A self-regulating brake on metabolism

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Topic 6 — ATP: Energy Currency of the Cell

Easy

Q	A
What does ATP stand for?	Adenosine Triphosphate
Why is ATP called the "energy currency" of the cell?	It stores and releases energy that can be used for all cellular functions, just as money is used for all economic transactions
What are the three components of an ATP molecule?	Adenine (nitrogen base), Ribose (5-carbon sugar), and three Phosphate groups
What is ADP?	Adenosine Diphosphate — ATP after one phosphate group is removed

What is AMP?	Adenosine Monophosphate — adenosine with only one phosphate group
What are "high-energy bonds" in ATP?	The covalent bonds between phosphate groups that store large amounts of energy (represented by wavy lines)
How is energy released from ATP?	By hydrolysis — adding water to break the bond between the second and third phosphate groups

Medium

Q	A
Write the ATP–ADP cycle equation.	ATP + H_2O → ADP + P_i + energy (release); ADP + P_i + energy → ATP (recharge)
What is phosphorylation?	The process of adding a phosphate group to ADP to reform ATP
How many moles of ATP do human cells hydrolyze per day?	100–150 moles
How many ATP molecules are produced per glucose in aerobic respiration?	36 ATP
Name two processes that recharge ADP back to ATP.	Cellular respiration (catabolism) and photosynthesis (light energy)
What percentage of the body's ATP does the brain use?	25%

Hard

Q	A
Why is it advantageous to use one universal energy currency (ATP) rather than having different energy molecules for different reactions?	It standardizes energy transfer — any reaction that needs energy can use ATP regardless of where the energy originally came from (sunlight, glucose, fat). This makes metabolic pathways interoperable and efficient
The same ATP molecule can be recycled hundreds of times per day. Why is this possible?	Hydrolysis of ATP → ADP is reversed by phosphorylation using energy from respiration/photosynthesis. The molecule itself is not destroyed — only the phosphate bond is broken and reformed

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Topic 7 — Photosynthesis

Easy

Q	A
Write the overall equation for photosynthesis.	$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$ (with light and chlorophyll)
Where does photosynthesis take place?	In chloroplasts (in plants, algae, and cyanobacteria)
What are the two phases of photosynthesis?	Light-dependent reactions and light-independent reactions (Calvin cycle)
Where do light-dependent reactions occur?	In the thylakoid membranes (grana) of the chloroplast
Where does the Calvin cycle occur?	In the stroma of the chloroplast
What is photolysis?	The splitting of water molecules by light energy in Photosystem II
What gas is produced as a by-product of photosynthesis?	Oxygen (O_2)
Name the two products of light-dependent reactions used in the Calvin cycle.	ATP and NADPH

Medium

Q	A
What are the products of the light-dependent reactions?	ATP, NADPH, and oxygen
What molecule captures CO_2 in the Calvin cycle?	RuBP (Ribulose Bisphosphate — 5-carbon molecule)
What is the first stable product after CO_2 enters the Calvin cycle?	3-PGA (3-phosphoglycerate) — a 3-carbon molecule
What is G3P?	Glyceraldehyde-3-phosphate — the 3-carbon sugar produced in the Calvin cycle; building block for glucose
Why are dark reactions also called light-independent reactions?	They don't directly use light; they use ATP and NADPH made by the light reactions. They can occur in the dark if ATP and NADPH are provided
Name the three limiting factors of photosynthesis.	Light intensity, CO_2 concentration, and temperature
What is the optimum temperature for photosynthesis?	25°C
What are the two photosystems involved in light-dependent reactions?	Photosystem I and Photosystem II
What does Photosystem II absorb to release electrons?	Light

Hard

Q	A
In the light-dependent reactions, why must water be split?	When light hits Photosystem II, electrons are ejected from chlorophyll. These must be replaced. Water is split (photolysis) to release electrons that replace those lost by chlorophyll — also releasing O_2 as a by-product
How is ATP synthesized during the light-dependent reactions?	Electrons from Photosystem II pass through the Electron Transport Chain, releasing energy that is used for ATP synthesis (from ADP + P_i)
How many turns of the Calvin cycle are needed to produce one glucose?	3 turns of the cycle fix 3 CO_2 to produce enough G3P to regenerate RuBP and output 1 G3P net; 6 turns are needed to produce 1 glucose molecule

Topic 8 — Respiration

Easy

Q	A
Write the overall equation for aerobic respiration.	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 36\ ATP$
What are the two types of respiration?	Aerobic and anaerobic
What is the first stage of both aerobic and anaerobic respiration?	Glycolysis
Where does glycolysis occur?	Cytoplasm
What is the equation for glycolysis?	Glucose → 2 Pyruvic Acid + 2 ATP + 2 NADH
What is anaerobic respiration also called?	Fermentation
Name the two types of anaerobic fermentation.	Alcoholic fermentation and lactic acid fermentation
What are the products of alcoholic fermentation?	Ethyl alcohol and carbon dioxide
What is the product of lactic acid fermentation?	Lactic acid

How many ATP are produced by anaerobic respiration?	2 ATP
How many ATP are produced by aerobic respiration?	36 ATP

Medium

Q	A
Name the four stages of aerobic respiration.	Glycolysis, Formation of Acetyl CoA, Krebs cycle, Electron Transport Chain
Where does the Krebs cycle take place?	Mitochondrial matrix
Where does the Electron Transport Chain (ETC) take place?	Inner membrane of mitochondria
What is the role of oxygen in the ETC?	It is the final electron acceptor; it combines with hydrogen ions to form water
What does the equation 2 Pyruvic Acid + 2 CoA → 2 Acetyl CoA + 2 CO_2 + 2 NADH represent?	Formation of Acetyl CoA — the transition step between glycolysis and the Krebs cycle
What is the Krebs cycle equation?	2 Acetyl CoA → 4CO_2 + 2 ATP + 6 NADH + 2 $FADH_2$
Which organisms carry out alcoholic fermentation?	Yeast and some bacteria
When does your body produce lactic acid?	During vigorous exercise when muscles lack sufficient oxygen
Name 4 uses of ATP energy in the body.	Muscle contraction, nerve impulse transmission, active transport, growth and repair of cells

Hard

Q	A
Why is aerobic respiration 18 times more efficient than anaerobic respiration?	Aerobic: 36 ATP per glucose. Anaerobic: 2 ATP per glucose. The extra 34 ATP come from the Krebs cycle and ETC which require oxygen to run. Without O_2, only glycolysis runs
Explain how glycolysis is common to both aerobic and anaerobic respiration, but the pathways diverge at pyruvate.	Glycolysis splits glucose to 2 pyruvate in the cytoplasm regardless of oxygen. In aerobic conditions, pyruvate enters the mitochondria and continues through Krebs + ETC. In anaerobic conditions, pyruvate is instead converted to alcohol or lactic acid in the cytoplasm
How are photosynthesis and aerobic respiration reverse processes of each other?	Photosynthesis: 6CO_2 + 12H_2O + energy → glucose + O_2; Respiration: glucose + 6O_2 → 6CO_2 + 6H_2O + energy. The reactants of one are the products of the other

Mixed Review

These cards shuffle across topic boundaries — the kind of thinking exams require.

Q	A
An enzyme is both a protein and a catalyst. How do these two properties relate?	Being a protein gives it the 3D shape (active site) that makes it a specific catalyst; the protein's shape determines which substrate it can bind and what reaction it speeds up
ATP is produced by catabolism and used in anabolism. Explain the link.	Catabolism breaks down molecules, releasing energy that is captured to recharge ADP → ATP. Anabolism uses the ATP energy to build complex molecules. ATP is the transfer agent between the two
You eat bread (starch). Trace the metabolic journey to muscle movement.	Starch is hydrolyzed (catabolism) → glucose → glycolysis (cytoplasm) → Acetyl CoA → Krebs cycle → ETC → 36 ATP → muscle proteins use ATP for contraction
Why does your body temperature of 37°C matter for your enzymes?	37°C is the optimum temperature for most human enzymes. Deviating significantly (fever >40°C or hypothermia) impairs enzyme function and can be life-threatening through denaturation

Photosynthesis and respiration are described as "interlinked." Explain.	The O_2 released by photosynthesis is used by respiration; the CO_2 released by respiration is used by photosynthesis. The glucose made by photosynthesis is the fuel for respiration. They are the two halves of the Earth's carbon-oxygen cycle
A drug inhibits Photosystem II. What happens to ATP production in the leaf?	Light-dependent reactions would stop → no ATP or NADPH produced → Calvin cycle cannot run → photosynthesis halts completely
Why does the brain suffer first when oxygen is cut off (e.g., in drowning)?	Brain cells depend almost entirely on aerobic respiration (36 ATP/glucose). Without O_2, ETC stops, ATP production collapses. The brain uses 25% of all ATP and has no significant anaerobic capacity — it exhausts its energy stores in minutes
How does the concept of the metabolic pathway relate to enzyme specificity?	A metabolic pathway is a series of reactions where each step is catalyzed by a specific enzyme. Because each enzyme is specific to one reaction, the pathway is like an assembly line — each worker does only their job, in sequence, reliably
Why is fermentation by bacteria useful to humans?	Bacteria use lactic acid fermentation to make yogurt and cheese; yeast use alcoholic fermentation to make bread, beer, and wine. Human food technology has exploited anaerobic metabolism for thousands of years
How does lowering temperature during surgery help the patient?	Cold slows enzyme activity in brain and other cells, dramatically reducing their ATP demand. This allows surgeons to temporarily reduce blood flow to the brain without causing brain damage — the cells are in metabolic "slow motion"